Tribhuvan University
Institute of Engineering
Purwanchal Campus
पूर्वाञ्चल क्याम्पस

UML Design Report

Project: Decentralized Smart Power Grid

Department of Computer & Electronics Engineering

Course	Software Engineering (Practical)
Course code	ENCT 352
Team	440 V
Submitted on	July 21, 2026

Team Members

S.N.	Name	Roll / ID
1	PRATIMA CHAUDHARY	PUR080BCT064
2	RABIN KATTEL	PUR080BCT067
3	SABIN BASNET	PUR080BCT077
4	SARBESH TIMSINA	PUR080BCT082

Submitted to: Asst. Prof. Binay Lal Shrestha

Contents

1 Introduction

1.1 Purpose

This report presents the Unified Modeling Language (UML) design for the proposed **Decentralized Smart Power Grid** system. The report includes the Use Case, Sequence, Activity, and Class diagrams, which collectively illustrate the functional behavior, interactions, workflows, and structural design of the system. These diagrams provide a comprehensive representation of how consumers, administrators, smart meters, and backend services interact to support secure energy monitoring, data management, anomaly detection, and real-time system monitoring.

1.2 Scope

The UML diagrams describe the runtime behavior and structural organization of the Decentralized Smart Power Grid system. They cover user interactions, operational workflows, object collaborations, and class relationships involved in smart meter data collection, energy consumption monitoring, anomaly detection, and dashboard management. The implementation details of machine learning models, blockchain infrastructure, hardware development, and database optimization are outside the scope of this report.

1.3 Definitions and Abbreviations

Term	**Meaning**
UML	Unified Modeling Language
Use Case Diagram	Represents interactions between actors and the system
Sequence Diagram	Illustrates the order of interactions among system objects
Activity Diagram	Models the workflow and execution of system processes
Class Diagram	Represents the static structure of classes and their relationships
IoT	Internet of Things
ESP32	Microcontroller used as the smart meter node
API	Application Programming Interface
ML	Machine Learning
Blockchain	Distributed ledger technology for secure data recording
Telemetry	Real-time energy usage data transmitted by the smart meter
Consumer	Registered electricity user of the system
Administrator	Utility operator responsible for monitoring the system

2 Project Domain Analysis

2.1 Domain Description

Electrical power distribution systems are responsible for delivering electricity from utility providers to consumers while continuously monitoring energy consumption and ensuring reliable grid operation. Traditional systems rely on centralized metering infrastructure for collecting consumption data and managing system operations. The proposed project introduces a decentralized approach that integrates IoT smart meters, blockchain technology, and machine learning to provide secure data management, transparent monitoring, anomaly detection, and real-time system supervision.

2.2 Existing System

In conventional power distribution systems, smart meters or electricity meters transmit consumption data to a centralized utility database for storage and processing. System monitoring and energy management are primarily controlled through centralized infrastructure, creating a single point of failure and limiting transparency in data management. Consumers typically have limited access to real-time information regarding their energy consumption and system status.

2.3 Proposed System

The proposed Decentralized Smart Power Grid continuously collects electricity consumption data from ESP32-based smart meter nodes. The backend validates incoming telemetry, securely records system transactions using blockchain technology, and analyzes consumption patterns through machine learning to detect abnormal usage and possible power theft. The processed information is stored securely and presented through a web dashboard, enabling consumers and administrators to monitor energy consumption, historical records, system performance, and alerts in real time.

3 Use Case Diagram Design

3.1 Actors

The Decentralized Smart Power Grid system involves three primary actors:

- Consumer
- Grid Administrator
- Smart Meter (ESP32)

The consumer monitors electricity usage, manages prepaid balance, and receives notifications. The administrator supervises system operations, monitors alerts, and manages consumers. The smart meter continuously sends energy consumption data to the backend.

3.2 Use Case Diagram

The use case diagram illustrates the interactions between external actors and the major functionalities provided by the Decentralized Smart Power Grid System.

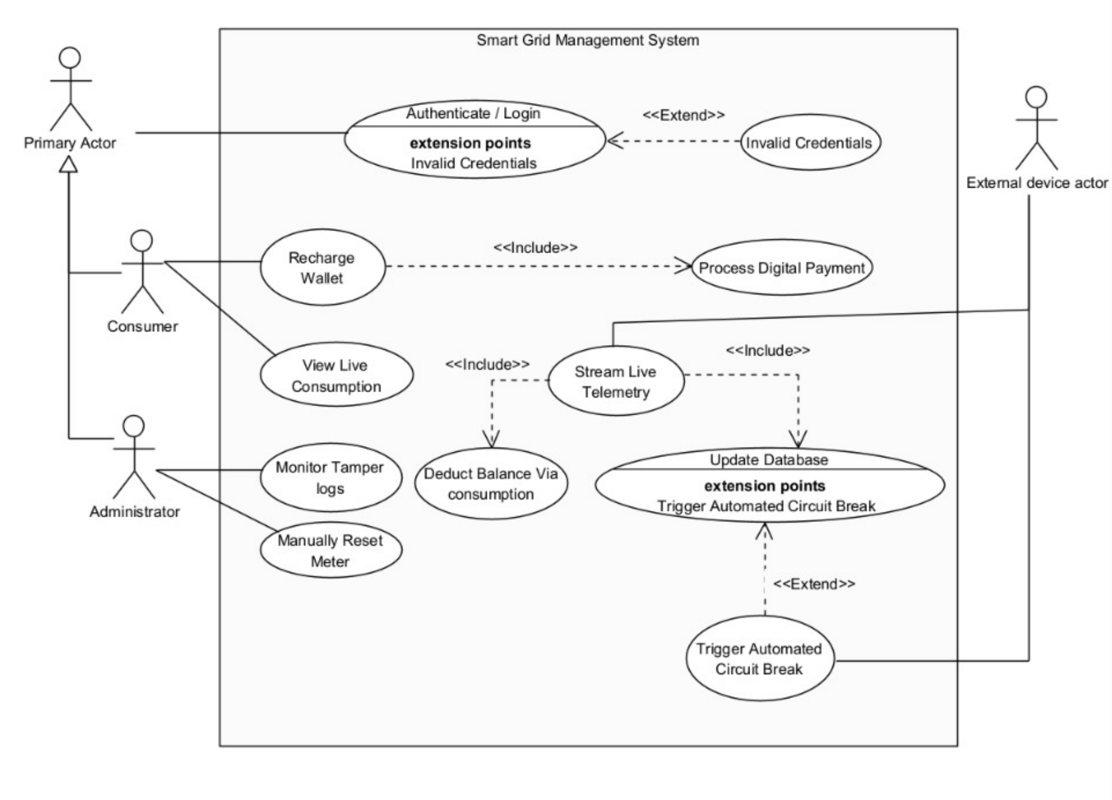

Figure 1: Use Case Diagram of the Decentralized Smart Power Grid System

3.3 Use Case Description

The diagram shows how consumers interact with the system to monitor energy consumption, recharge prepaid balances, and receive notifications, while administrators monitor grid status and manage consumers. The smart meter supplies real-time telemetry to support automated billing and monitoring.

4 Sequence Diagram Design

4.1 Scenario Description

The sequence diagram illustrates the interaction between the Smart Meter, Backend Server, Blockchain Smart Contract, Database, and Consumer during the prepaid billing process.

4.2 Sequence Diagram

Figure 2 illustrates the chronological sequence of messages exchanged among system components.

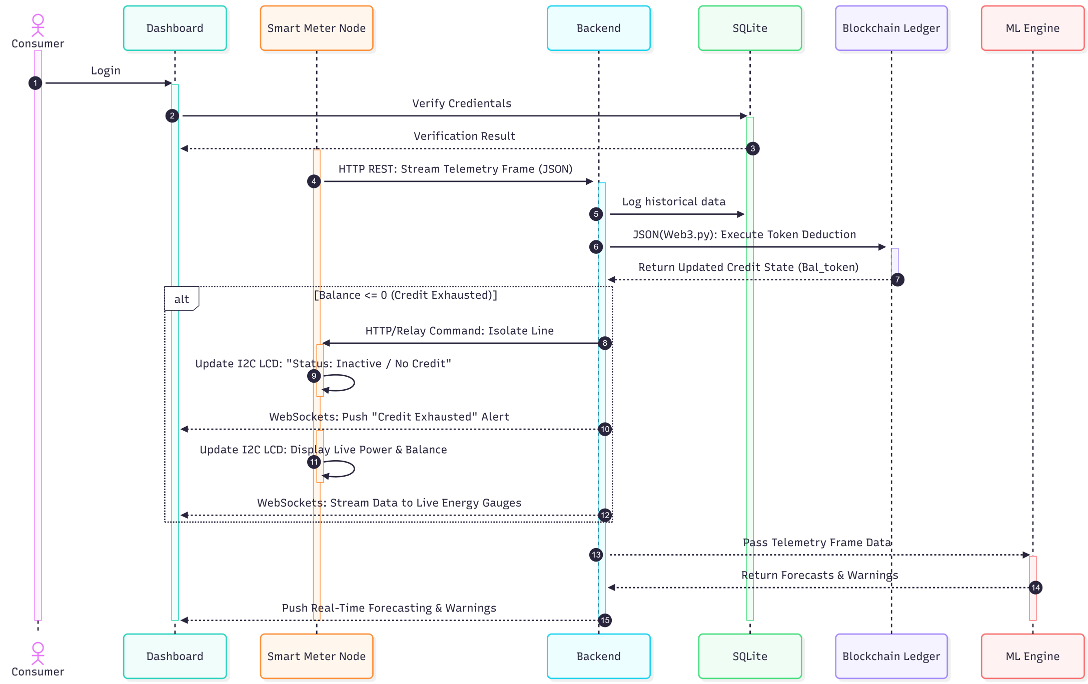

Figure 2: Sequence Diagram of the Decentralized Smart Power Grid System

4.3 Interaction Description

The smart meter periodically transmits electricity consumption data to the backend. After validating the telemetry, the backend invokes the blockchain smart contract to deduct prepaid credit. The updated balance is stored in the database, and the dashboard is refreshed to provide real-time information to consumers and administrators.

5 Activity Diagram Design

5.1 Workflow Description

The activity diagram models the operational workflow of the Decentralized Smart Power Grid System from meter data collection to dashboard updates.

5.2 Activity Diagram

Figure 3 presents the activity flow.

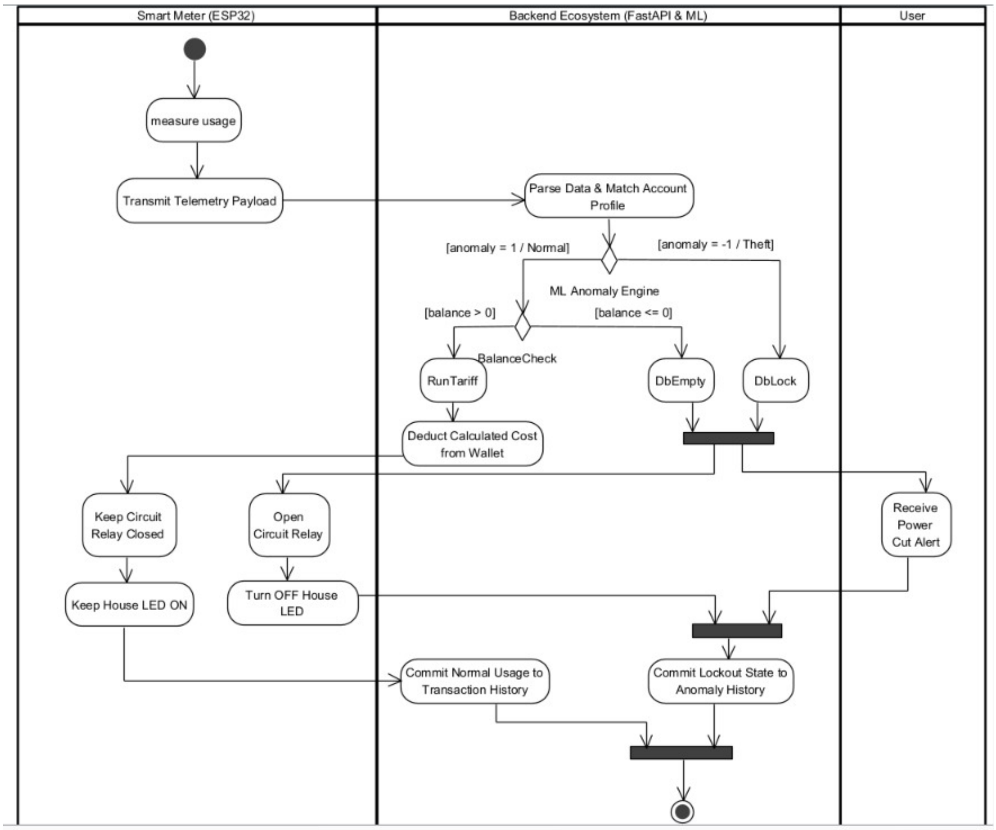

Figure 3: Activity Diagram of the Decentralized Smart Power Grid System

5.3 Workflow Explanation

The workflow begins with collecting energy consumption data from the smart meter. The backend validates the incoming telemetry, executes blockchain-based prepaid billing, performs anomaly detection, updates system records, and finally displays the latest information through the dashboard.

6 Class Diagram Design

6.1 Class Identification

The class diagram represents the static structure of the Decentralized Smart Power Grid System by identifying the primary classes, their attributes, operations, and relationships.

6.2 Class Diagram

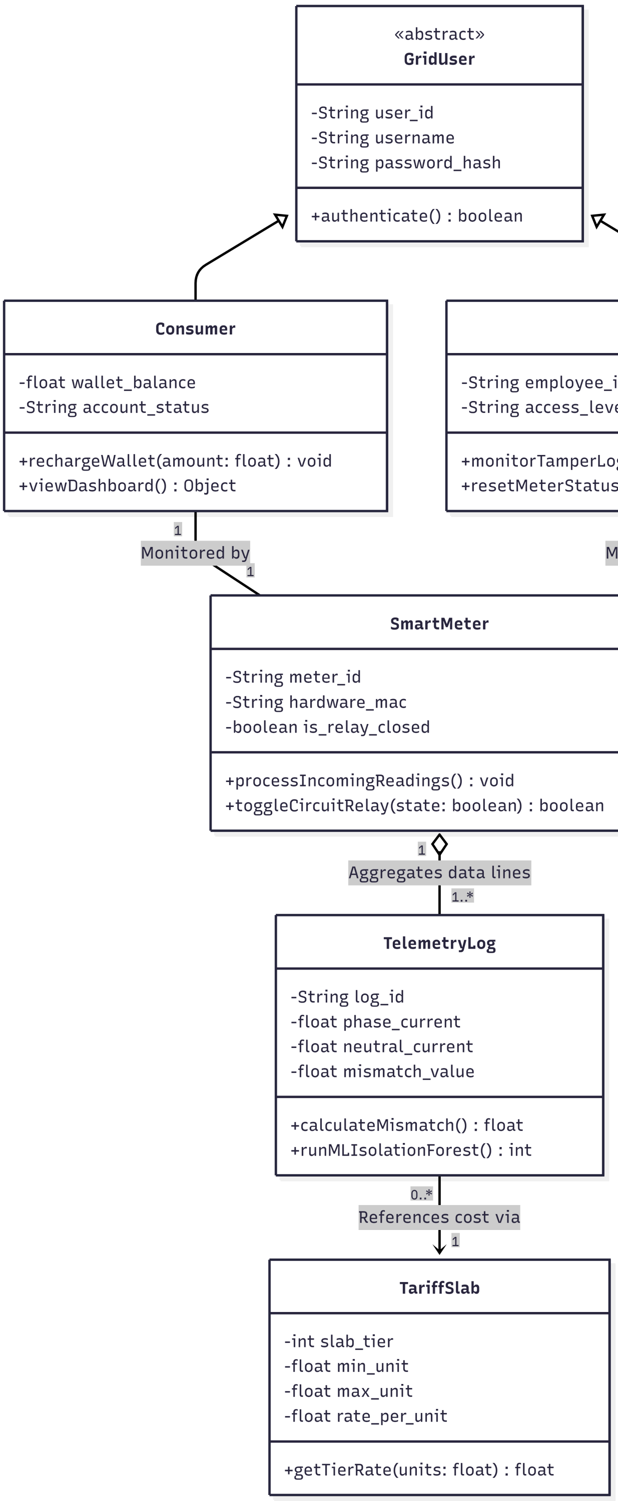

Figure 4: Class Diagram of the Decentralized Smart Power Grid System

6.3 Class Description

The principal classes include Consumer, Smart Meter, Backend Server, Smart Contract, Database, and Administrator. These classes collaborate to collect energy consumption data, process prepaid billing through blockchain technology, detect abnormal usage, and provide real-time monitoring facilities.

7 Requirements Derived from the UML Design

7.1 Functional Requirements

ID	Functional Requirement	Priority
FR-1	The system shall receive real-time energy consumption data from ESP32-based smart meter nodes.	High
FR-2	The system shall validate and process incoming telemetry data before storing and analyzing it.	High
FR-3	The system shall securely record energy consumption and system transactions using blockchain technology.	High
FR-4	The system shall detect abnormal energy consumption patterns and possible power theft using machine learning techniques.	High
FR-5	The system shall maintain energy consumption records, user information, and system logs.	Medium
FR-6	The system shall provide dashboards for consumers and administrators to monitor energy usage, system status, and alerts in real time.	Medium

7.2 Non-functional Requirements

ID	Non-functional Requirement	Category
NFR-1	The system should process incoming telemetry data with minimal delay to support near real-time monitoring.	Performance
NFR-2	The dashboard should provide an intuitive and user-friendly interface for monitoring energy consumption, system status, and alerts.	Usability
NFR-3	The system should ensure the integrity and security of energy data and blockchain transaction records.	Security
NFR-4	The system should maintain reliable communication between smart meter nodes, backend services, databases, and blockchain components.	Reliability

8 Conclusion

This report presented the Unified Modeling Language (UML) design for the proposed Decentralized Smart Power Grid system. The Use Case, Sequence, Activity, and Class diagrams collectively describe the functional behavior, system workflows, object interactions, and structural organization of the proposed solution.

The UML diagrams provide a comprehensive representation of the system by illustrating user interactions, operational processes, communication between system components, and relationships among the primary classes. These models establish a clear foundation for the subsequent phases of software design, implementation, integration, and testing while ensuring that the system requirements and architecture are well defined and understood.

9 References

1. IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
2. Smith, J., & Kumar, A. (2022). IoT-Based Smart Energy Meter for Real-Time Monitoring and Automated Billing. International Journal of Smart Grid Technologies, 10(3), 150-158.
3. J. Jithish, B. Alangot, N. Mahalingam, and K. S. Yeo, "Distributed Anomaly Detection in Smart Grids: A Federated Learning-Based Approach," IEEE Access, vol. 11, pp. 9824-9838, 2023.